



OPEN SmartSport: crowd counting meets large language models for smart facility management

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Efficient management and optimal allocation of urban public sports facilities represents a critical challenge in modernizing public sports service governance. Current approaches to facility utilization assessment predominantly depend on manual counting and questionnaire surveys, which are constrained by low efficiency, limited coverage, and poor real-time responsiveness. Meanwhile, conventional management methods lack the capability to extract actionable insights from heterogeneous data sources, hindering intelligent decision-making. To address these limitations, this study introduces SmartSport, an intelligent management framework for public sports facilities that synergizes computer vision with large language models. The framework consists of two core modules. The CrowdVision module constructs a crowd counting network based on a lightweight visual state space model, leveraging selective state space mechanisms for global context modeling, cross-scan modules for spatial dependency capture, and multi-level feature pyramids for enhanced multi-scale perception. A point query mechanism is further incorporated to enable precise counting alongside crowd localization. The LLM-Advisor module employs large language models as the analytical engine, integrating crowd time-series data with geographic information, demographic distributions, and other contextual data in a structured manner. Through carefully designed prompt engineering, the model performs systematic analysis across three dimensions: usage pattern recognition, supply-demand gap diagnosis, and optimization recommendation generation, ultimately producing comprehensive management reports with problem attribution and actionable configuration suggestions. Experimental results on our constructed dataset demonstrate that the CrowdVision module achieves a counting accuracy of 93.8%, while recommendations generated by the LLM-Advisor module received a practicality score of 4.2 out of 5.0 based on blind evaluation by domain experts.

Keywords Public sports facilities, Crowd counting, Visual state space models, Large language models, Intelligent decision support

The implementation of national health initiatives and fitness strategies has significantly advanced the development of urban public sports facilities in many countries, establishing preliminary public fitness service systems covering both urban and rural areas. However, the rapid growth in facility quantity has not translated proportionally into improved service efficiency. Structural supply-demand imbalances persist across cities: basketball courts and badminton halls in popular areas operate under chronic overload, while fitness trails and outdoor exercise equipment in less accessible locations remain consistently underutilized. This mismatch reflects fundamental deficiencies in current facility management regarding demand identification accuracy and resource allocation rationality. From a public administration perspective, addressing this challenge requires establishing data-driven monitoring systems and intelligent decision-making mechanisms to transform public sports services from experience-based management to evidence-based governance.

Existing research on public sports facility management optimization exhibits two significant limitations, as illustrated in Fig 1. First, regarding utilization data collection, traditional methods struggle to balance efficiency, accuracy, and coverage. Manual counting is costly and yields only discrete temporal snapshots; questionnaire surveys are constrained by sample representativeness and recall bias; IC card or reservation systems apply only to enclosed fee-charging facilities and cannot address the larger proportion of open-access public fitness venues. Second, at the management analysis and decision-making level, current approaches lack sufficient intelligence. Statistical analysis can reveal spatiotemporal distribution patterns but fails to explain underlying mechanisms; geographic information systems can visualize layout conditions but cannot automatically identify configuration

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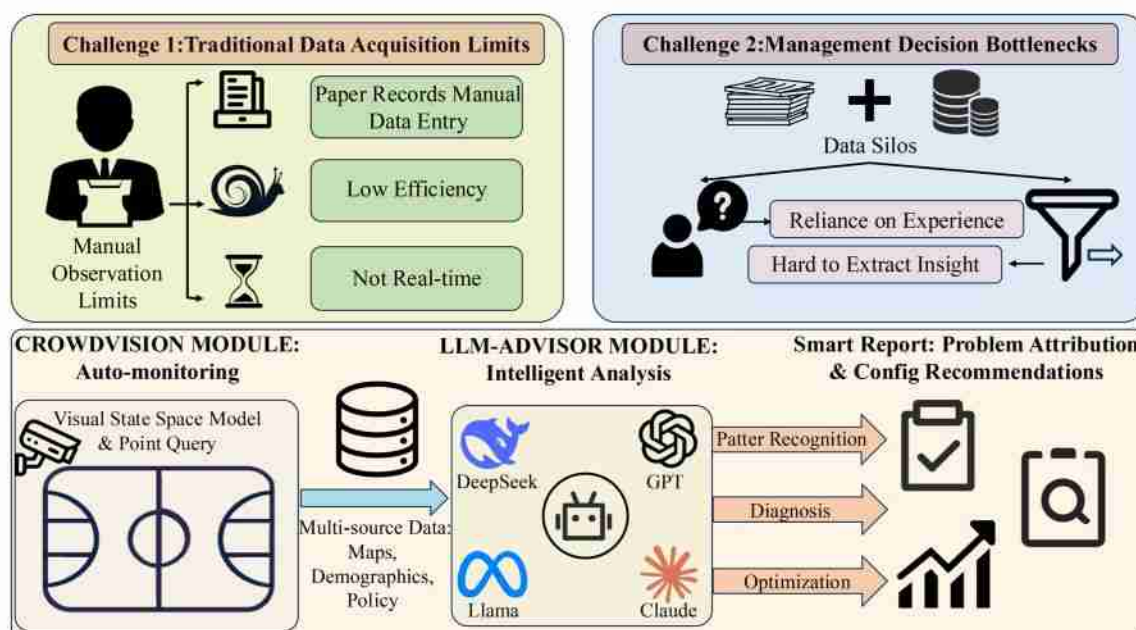


Fig. 1. Challenges in public sports facility management and overview of the SmartSport framework.

problems; operations research models can solve for theoretical optima but often disconnect from management realities. More critically, these methods operate in isolation without effective integration, requiring managers to navigate multiple systems and synthesize judgments independently, compromising both decision efficiency and quality.

To address these challenges, this study proposes SmartSport, an intelligent management framework for public sports facilities that integrates visual perception with large language models. The framework employs computer vision technology to solve data collection problems and large language models to address data analysis and decision generation, constructing a complete pipeline from automated monitoring to intelligent decision-making. SmartSport comprises two core modules: The **CrowdVision module** aims to achieve automated monitoring of facility usage. This module deploys network cameras at key facility locations and constructs a crowd counting network using a lightweight Visual State Space Model. Compared with traditional convolutional networks, state space models achieve global context modeling through selective state space mechanisms while maintaining linear computational complexity, making them better suited for sports facility scenarios characterized by wide fields of view and uneven crowd distributions. The network employs Cross-Scan Modules to perform feature scanning from four image directions to capture global spatial dependencies, and designs a multi-level feature pyramid structure to fuse high-level semantic information with low-level spatial details, enhancing perception capability for crowds at different scales. Additionally, a point query mechanism is introduced to model crowd counting as a decomposable point prediction process, providing precise counting while supporting crowd localization. The lightweight network design enables deployment on edge computing devices for real-time inference, with the module automatically capturing images at preset intervals, completing head counts, and uploading results to the management platform to form time-series usage data for each facility. The **LLM-Advisor module** aims to achieve intelligent generation of management recommendations. This module employs large language models as the core analysis engine, with a workflow divided into three stages: first, the data integration stage structurally organizes crowd time-series data output by the CrowdVision module with facility geographic location, surrounding population density, transportation accessibility, and supporting facilities to construct facility profiles; second, the analytical reasoning stage designs specialized prompt templates for facility management scenarios based on prompt engineering techniques, guiding the large language model to perform systematic analysis across three dimensions including usage pattern recognition (such as peak hours and periodic patterns), supply-demand diagnosis (such as facility overload or underutilization), and optimization recommendation generation (such as operating hour adjustments and new facility siting); finally, the report generation stage integrates analysis results into natural language management reports containing facility usage descriptions, problem attribution analysis, and actionable configuration optimization recommendations. Experimental results on our constructed dataset demonstrate the effectiveness of the proposed SmartSport framework. The CrowdVision module achieves a crowd counting accuracy of 93.8% on test sets containing outdoor basketball courts, fitness trails, soccer fields, and other scenario types; management recommendations generated by the LLM-Advisor module received a practicality rating of 4.2 out of 5 points through blind evaluation by 12 domain experts.

The contributions of this study are as follows: (1) proposing SmartSport, a public sports facility management framework integrating computer vision and large language models, achieving a complete pipeline from automated facility usage monitoring to intelligent management recommendation generation; (2) designing CrowdVision, a crowd perception module based on density estimation, optimized for open-access sports facility

scenarios and supporting low-cost deployment with continuous monitoring; (3) developing LLM-Advisor, an intelligent analysis module based on large language models that integrates multi-source heterogeneous data through prompt engineering techniques to automatically generate management reports containing problem diagnosis and configuration recommendations.

Related work

Public sports facility management and utilization assessment

Spatial layout optimization and utilization assessment of public sports facilities are core issues in sports public service research. Geographic Information Systems (GIS) and spatial analysis methods have been widely applied in this field. Researchers have utilized GIS spatial analysis tools based on Point of Interest (POI) data to study the spatial distribution characteristics of sports venues¹, or examined the spatial clustering characteristics and accessibility of public sports facilities at street and grid scales². Regarding facility layout equity assessment, existing studies have proposed spatial equity measurement methods based on multi-source urban data and GIS network analysis³, or employed buffer analysis and Lorenz curves to reveal facility distribution patterns⁴. While these studies have made significant progress in spatial layout analysis, they exhibit notable deficiencies in dynamic monitoring of actual facility usage. Current research predominantly relies on potential demand estimation based on demographic data or subjective reports from questionnaire surveys, making it difficult to obtain continuous, objective data on facility usage. Some studies have attempted to introduce intelligent monitoring approaches, with IoT sensors being applied in smart sports stadiums⁵. However, these solutions primarily target large enclosed venues with high deployment costs, making them difficult to extend to the vast number of open public fitness facilities. The CrowdVision module proposed in this study provides a feasible solution for low-cost, continuous monitoring of open sports facilities through lightweight visual counting technology.

Crowd counting and density estimation techniques

Crowd counting, which aims to estimate the number of people from images, is a classic task in computer vision that has been systematically reviewed in numerous surveys⁶. CNN-based methods represent the current mainstream framework, with key techniques including visual attention mechanisms, dilated convolutions, and spatial pyramid pooling, which have significantly improved density map estimation quality. Representative works such as SC2Net⁷ employ pyramid dilated convolutions for scale-aware counting, HRANet⁸ proposes hierarchical region-aware networks for handling complex scenes, and BL⁹ enhances counting robustness through Bayesian loss functions. Transformer architectures compensate for CNN's limitations in global modeling through self-attention mechanisms: CCTrans¹⁰ first introduced Transformers to crowd counting, TransCrowd¹¹ achieved pure Transformer-based weakly supervised counting. State Space Models (SSM) represent an emerging paradigm in visual modeling. Vision Mamba¹² first introduced bidirectional SSM into visual representation learning, achieving global context modeling while maintaining linear complexity, with inference efficiency significantly superior to Transformers. VMamba¹³ further explored scanning strategies adapted to 2D visual data, while LocalMamba¹⁴ focuses on local dependency modeling. However, SSM applications in crowd counting remain in the exploratory stage. The CrowdVision module in this study introduces visual state space models to sports facility scenarios, combining point query mechanisms to achieve efficient and accurate crowd counting and localization.

Large language models in urban management

Large Language Models (LLMs), with their powerful natural language understanding and generation capabilities, are becoming new intelligent engines for urban management and decision support¹⁵, with primary applications in planning management, public services, and transportation mobility. In urban activity management, UrbanLLM¹⁶ decomposes urban queries into subtasks and invokes spatiotemporal AI models to generate comprehensive responses, significantly outperforming general LLMs on complex planning problems. Participatory urban planning research¹⁷ generates planning schemes meeting diverse needs through LLM multi-agent collaboration frameworks. PlanGPT¹⁸ integrates local databases to support policy drafting and zoning assessment, while agents such as TrafficGPT¹⁹ enable interactive analysis and report generation for traffic data. The development of multimodal LLMs has further expanded application boundaries, with vision-language models demonstrating capabilities for automated analysis and report writing in fields such as medical report generation²⁰. Regarding prompt engineering, research has found that structured prompt formats can significantly improve LLM response quality²¹. InsightLens²² adopts the ReAct paradigm for data analysis planning and execution, and prompt engineering techniques for time-series data can effectively perform pattern recognition and anomaly detection^{23,24}. Although LLMs are widely applied in urban management, existing work primarily focuses on macro-level planning. The LLM-Advisor module in this study applies LLMs to sports facility management scenarios, integrating visual counting data with multi-source geographic information through specialized prompt templates to achieve automated generation of usage pattern recognition, supply-demand diagnosis, and configuration recommendations. More broadly, LLMs have shown growing potential in time series and spatio-temporal data analysis. Cross-modality alignment techniques have been explored to bridge numerical time series and LLM token spaces^{25,26}, while continuous learning frameworks address catastrophic forgetting in streaming spatio-temporal prediction²⁷. Recent reviews further systematize spatio-temporal foundation model pipelines²⁸, and efficient dataset condensation methods reduce the cost of large-scale temporal data management²⁹. These advances provide broader methodological context for our LLM-Advisor module, which integrates crowd time-series data with multi-source contextual information for facility management decision-making.

Methodology

SmartSport framework overview

SmartSport is an intelligent management framework for public sports facilities that integrates computer vision and large language models, aiming to achieve an end-to-end pipeline from automated facility usage monitoring to intelligent management recommendation generation. As illustrated in Fig. 2, the framework consists of two core modules: the CrowdVision module is responsible for real-time monitoring of facility usage, while the LLM-Advisor module is responsible for generating management recommendations based on multi-source data.

The overall workflow can be formally expressed as:

$$\mathcal{M} = \text{LLM-Advisor}(\text{DataIntegration}(\text{CrowdVision}(\mathcal{I}_t), \mathcal{D}_{\text{multi}})) \quad (1)$$

where $\mathcal{I}_t$ represents the video frame sequence at time t , $\mathcal{D}_{\text{multi}}$ represents multi-source data including geographic information and population distribution, and $\mathcal{M}$ is the output management report.

CrowdVision module

The CrowdVision module aims to achieve automated monitoring of facility usage, with core challenges including handling wide fields of view, uneven crowd distributions, and complex lighting variations in open-access scenarios. To address these challenges, we propose a crowd counting network based on a lightweight visual state space model, which achieves global context modeling through selective state space mechanisms while maintaining linear computational complexity, making it better suited for the characteristics of sports facility scenarios.

Lightweight visual state space network architecture

Considering the resource constraints of edge computing devices and real-time processing requirements, we adopt the lightweight ResNet-18 as the feature extraction backbone. Compared to deeper residual networks, ResNet-18 has fewer parameters and lower computational complexity, making it more suitable for deployment on resource-constrained edge devices, while its feature extraction capability is sufficient for crowd counting tasks. Given an input image $I \in \mathbb{R}^{H \times W \times 3}$, the backbone network progressively extracts multi-scale feature representations through four residual stages:

$$F^{(l)} = \text{ResBlock}^{(l)}(F^{(l-1)}) \in \mathbb{R}^{H/2^{l+1} \times W/2^{l+1} \times C_l} \quad (2)$$

where $l \in \{1, 2, 3, 4\}$, and $C_l \in \{64, 128, 256, 512\}$ correspond to the feature dimensions at each level. To accommodate the need for wide-range spatial perception in crowd counting tasks, we add dilated convolution

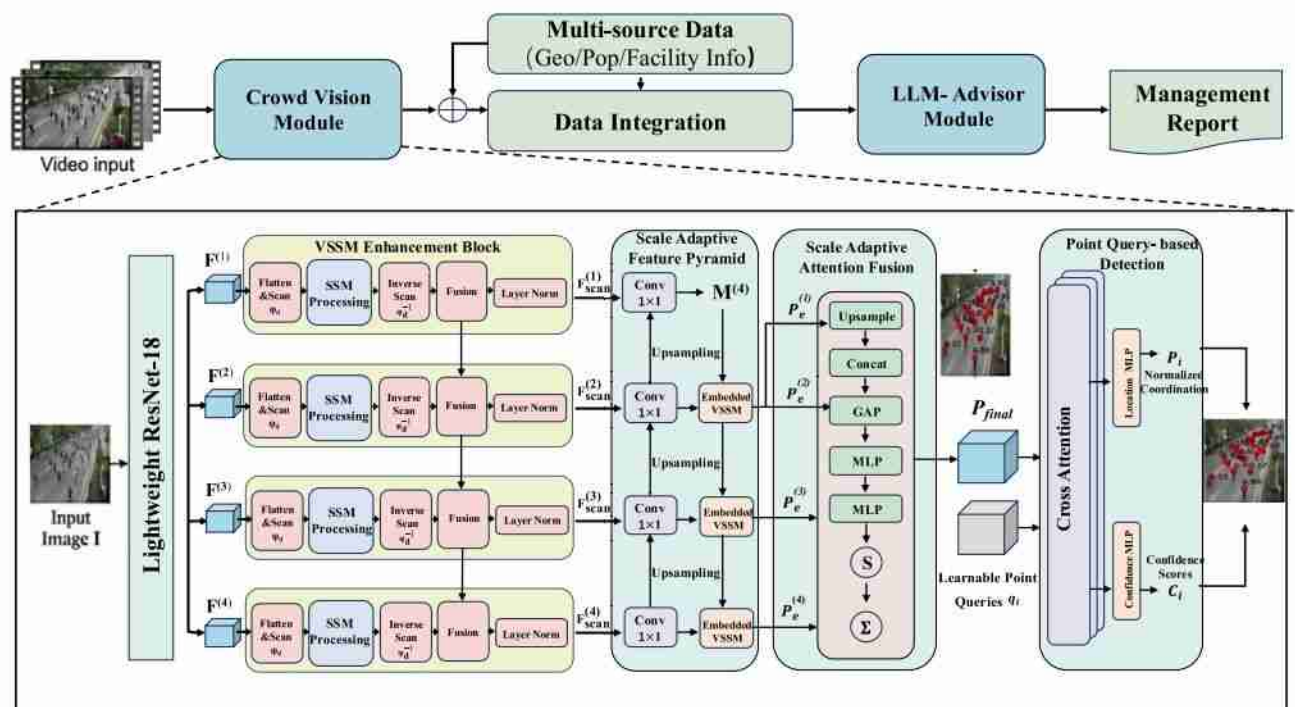


Fig. 2. Overview of the SmartSport framework (A) and detailed architecture of the CrowdVision module (B), illustrating the end-to-end pipeline from video input through VSSM-enhanced feature extraction, scale-adaptive feature pyramid, and point query-based detection to generate crowd counts and localization results for subsequent LLM-based management report generation.

modules after each residual block, enhancing the network's ability to capture dispersed crowds by expanding the receptive field without significantly increasing computational overhead.

Traditional convolutional networks suffer from receptive field limitations when handling long-range spatial dependencies, while Transformers can achieve global modeling but have quadratic computational complexity, making it difficult to meet real-time processing requirements. To address this, we introduce the Visual State Space Module (VSSM), which is based on state space theory and achieves global context modeling while maintaining linear complexity through recursive state update mechanisms. Unlike standard state space models that use fixed parameters, we introduce a selective mechanism that allows parameters to dynamically adjust based on input content. This mechanism enables the model to perceive local density features, enhancing information propagation in dense regions to model complex spatial relationships, while simplifying state updates in sparse regions to improve computational efficiency. For a 2D feature map $F \in \mathbb{R}^{H \times W \times D}$, VSSM first reshapes it into sequence form $X = \text{Flatten}(F) \in \mathbb{R}^{HW \times D}$. The recurrence form of the selective state space model is:

$$h_t = A_t h_{t-1} + B_t x_t, \quad y_t = C_t h_t + D x_t \quad (3)$$

where $h_t \in \mathbb{R}^D$ represents the hidden state vector, and x_t and y_t are the input and output respectively. The state transition matrix A_t , input projection matrix B_t , and output projection matrix C_t are dynamically generated from input features:

$$A_t = \text{softplus}(\Delta_A \cdot W_A x_t + b_A), \quad B_t = W_B x_t, \quad C_t = W_C x_t \quad (4)$$

where Δ_A is a learnable time step parameter controlling the rate of state updates, W_A, W_B, W_C are linear projection weights, b_A is a bias term, and the softplus activation function ensures numerical stability of the state transition matrix. The output sequence $Y = [y_1, y_2, \dots, y_T]$ is the processing result of the VSSM module.

Since 2D images have spatial structure, a single serialization direction would lose important spatial relationships. To fully capture omnidirectional spatial dependencies, we design a cross-directional scanning strategy that serializes the feature map from four main directions. Given a feature map $F \in \mathbb{R}^{H \times W \times D}$, we define the scanning function $\phi_d: \mathbb{R}^{H \times W \times D} \rightarrow \mathbb{R}^{HW \times D}$, where $d \in \{1, 2, 3, 4\}$ corresponds to four scanning modes: horizontal forward, horizontal backward, vertical forward, and vertical backward. Taking horizontal forward scanning as an example, feature elements are arranged in row-major order as $\phi_1(F)_{(i-1)W+j} = F_{i,j}$, where $i \in \{1, \dots, H\}$ and $j \in \{1, \dots, W\}$ are spatial coordinate indices. The sequence generated by each scanning direction is independently processed through VSSM to obtain output $Y^{(d)} = \text{VSSM}(\phi_d(F))$. Subsequently, the output sequences are reshaped back to the original spatial dimensions through inverse mapping functions ϕ_d^{-1} , and feature fusion is performed using learnable weights:

$$F_{\text{scan}} = \text{LayerNorm} \left(\sum_{d=1}^4 W_d \cdot \phi_d^{-1}(Y^{(d)}) \right) \quad (5)$$

where $W_d \in \mathbb{R}^{D \times D}$ is the fusion weight matrix for each direction, and LayerNorm is used to stabilize the training process. This multi-directional scanning strategy ensures that each spatial position can receive contextual information from both horizontal and vertical directions, effectively capturing the spatial correlation patterns of crowd distributions. The VSSM module processes the features $\{F^{(l)}\}_{l=1}^4$ output by each level of the backbone network separately, obtaining enhanced multi-scale features $\{F_{\text{scan}}^{(l)}\}_{l=1}^4$ as input for subsequent feature pyramid construction.

Multi-level feature pyramid and scale adaptation

Crowd distribution in sports facilities exhibits obvious multi-scale characteristics: people in foreground regions are larger and clearer, while those in background regions are smaller and more blurred. To effectively handle this scale variation, we construct a multi-level feature pyramid based on VSSM-enhanced features, integrating semantic information at different resolutions through a top-down feature fusion strategy.

Based on the VSSM-enhanced multi-scale features $\{F_{\text{scan}}^{(l)}\}_{l=1}^4$, feature pyramid construction is divided into two stages. First, channel compression is performed on features at each level through 1×1 convolution, unifying the feature dimensions of different levels to the same dimension space:

$$M^{(l)} = \text{Conv}_{1 \times 1}(F_{\text{scan}}^{(l)}) \in \mathbb{R}^{H/2^{l+1} \times W/2^{l+1} \times 128} \quad (6)$$

Then, a top-down fusion path is adopted to progressively transfer high-level semantic information to low-level features:

$$P^{(l)} = M^{(l)} + \text{Upsample}(P^{(l+1)}) \quad (7)$$

where the upsampling operation is implemented using bilinear interpolation, and $P^{(4)} = M^{(4)}$ serves as the starting point for recursion. To further enhance the feature representation capability at each pyramid level, we embed VSSM modules again after each level for global context enhancement:

$$P_{\text{enhanced}}^{(l)} = \text{VSSM}(P^{(l)}) + P^{(l)} \quad (8)$$

This hierarchical VSSM enhancement with residual connections provides implicit intermediate supervision, enabling each pyramid level to learn discriminative representations before the final fusion.

Considering that the contribution of different pyramid levels to crowd detection varies across scenarios, we introduce a scale-adaptive attention mechanism to dynamically adjust the fusion weights of each level. This mechanism first upsamples features from each level to a unified resolution and concatenates them, then extracts scene-level representations through global average pooling, and finally predicts the importance weights for each level through a multi-layer perceptron:

$$w = \text{Softmax}(\text{MLP}(\text{GAP}(\text{Concat}([P^{(1)}, \dots, P^{(4)}]))) \quad (9)$$

Based on the predicted weight vector $w = [w_1, w_2, w_3, w_4]^T$, the final multi-scale feature representation is the weighted fusion of enhanced features from each level:

$$P_{\text{final}} = \sum_{l=1}^4 w_l \cdot \text{Upsample}_{\text{to_base}}(P_{\text{enhanced}}^{(l)}) \quad (10)$$

where all features are upsampled to the base resolution $H/4 \times W/4$ to ensure subsequent processing is performed at a unified spatial scale. The fused feature P_{final} will serve as input to the point query mechanism, supporting subsequent crowd localization and counting.

Point query-driven crowd localization and counting

Traditional density estimation methods achieve crowd counting through density map regression, but have two key limitations: first, the supervision signal of density maps is sparse, and the superposition of Gaussian kernels easily leads to confusion of responses from adjacent targets; second, they only provide region-level counting results and cannot support precise individual localization. Meanwhile, Transformer-based methods suffer from quadratic computational complexity, making real-time edge deployment challenging. To address these issues, we propose a point query-driven crowd detection paradigm that models crowd counting as a combination of point detection tasks, simultaneously achieving counting and localization within a unified framework while maintaining computational efficiency.

Inspired by the end-to-end object detection method DETR, we design a set of learnable point query vectors $\{q_i\}_{i=1}^N \in \mathbb{R}^{128}$, where N is the preset maximum number of queries. Each query vector serves as a detection slot, specifically responsible for detecting one potential person location in the image. Unlike object queries in object detection that need to predict bounding boxes, point queries are more lightweight, requiring only prediction of 2D coordinates and existence confidence, significantly reducing prediction complexity. Query vectors interact with the feature pyramid through stacked cross-attention layers, progressively aggregating feature information related to target positions:

$$q_i^{(k+1)} = \text{CrossAttention}(q_i^{(k)}, P_{\text{final}}, P_{\text{final}}) + q_i^{(k)} \quad (11)$$

where $k \in \{0, 1, \dots, K-1\}$ represents the index of attention layers, $q_i^{(0)} = q_i$ is the initial query, and residual connections ensure stable gradient propagation. Through K layers of attention interaction, each query gradually focuses on a specific spatial region in the image, learning discriminative features related to the presence of people in that region.

After attention refinement, each query $q_i^{(K)}$ needs to output two types of prediction results: position coordinates and existence confidence. We adopt a dual-branch prediction head design, where both branches share the query representation but perform predictions independently:

$$p_i = \text{Sigmoid}(\text{MLP}_{\text{loc}}(q_i^{(K)})) \in [0, 1]^2, \quad c_i = \text{Sigmoid}(\text{MLP}_{\text{conf}}(q_i^{(K)})) \in [0, 1] \quad (12)$$

where the location prediction branch maps query vectors to normalized 2D coordinate space through a three-layer MLP, and the confidence prediction branch uses the same network structure to output existence probability. Based on prediction results, the headcount is obtained by counting queries with confidence exceeding threshold τ_{conf} : $\text{Count} = \sum_{i=1}^N 1(c_i > \tau_{\text{conf}})$, where the threshold value is determined by maximizing counting accuracy on the validation set. Meanwhile, the position coordinates predicted by queries with confidence exceeding the threshold constitute the point set representation of crowd spatial distribution: $\text{Locations} = \{(x_i, y_i) | c_i > \tau_{\text{conf}}\}$. This design unifies crowd counting and localization within the same prediction framework, providing both accurate total headcount statistics and individual-level location information, offering data support for regionalized refined management of sports facilities.

Multi-task joint optimization strategy

The CrowdVision module needs to simultaneously optimize two interrelated task objectives: crowd counting and location localization. Directly combining two types of loss functions may lead to training instability or one task dominating the optimization process. To address this, we design a multi-task joint optimization strategy with a three-level loss function, where each level targets a distinct optimization objective: (1) global counting consistency to ensure accurate total count estimation, (2) local position regression to achieve precise individual localization, and (3) confidence classification to enable reliable foreground-background discrimination. This hierarchical design provides complementary supervision signals at different granularities, ensuring balanced

learning of both counting and localization tasks while maintaining training stability. The strategy establishes correspondences between predictions and annotations through optimal bipartite matching, and employs adaptive weight mechanisms to balance the three loss components.

Since the number of point queries N is typically greater than the actual number of people $|\mathcal{G}|$ in the image (where $\mathcal{G}$ represents the ground truth annotation point set), we first need to establish one-to-one correspondences between predicted queries and ground truth annotations. Specifically, we use the Hungarian algorithm to solve the optimal bipartite matching problem:

$$\sigma^* = \arg \min_{\sigma \in \mathfrak{S}_N} \sum_{i=1}^{|\mathcal{G}|} \mathcal{C}(i, \sigma(i)) \quad (13)$$

where $\mathfrak{S}_N$ represents the set of all possible matching permutations. The matching cost function $\mathcal{C}(i, j)$ considers both position distance and confidence prediction:

$$\mathcal{C}(i, j) = \lambda_{\text{loc}} \|p_j - g_i\|_1 - \lambda_{\text{conf}} c_j \quad (14)$$

where g_i represents the normalized coordinates of the i -th ground truth annotation point, λ_{loc} and λ_{conf} are balancing coefficients, and the negative sign makes queries with higher confidence more likely to be selected for matching.

Based on the Hungarian matching result σ^* , we construct three levels of loss functions to supervise model training, each targeting different optimization objectives. The counting consistency loss directly constrains the consistency between predicted total count and ground truth count, optimizing counting precision from a global perspective:

$$\mathcal{L}_{\text{count}} = \left| \sum_{i=1}^N c_i - |\mathcal{G}| \right| \quad (15)$$

The location regression loss optimizes position prediction precision for successfully matched queries. Let $\mathcal{M}$ denote the set of query indices matched to ground truth points:

$$\mathcal{L}_{\text{loc}} = \sum_{i \in \mathcal{M}} \text{SmoothL1}(p_{\sigma^*(i)} - g_i) \quad (16)$$

where the SmoothL1 loss approximates L2 loss for small errors to provide smooth gradients, and degrades to L1 loss for large errors to maintain robustness.

The confidence classification loss guides each query to learn correct existence judgments, treating the matching problem as a binary classification task where matched queries have positive labels and unmatched queries have negative labels:

$$\mathcal{L}_{\text{conf}} = - \sum_{i=1}^N [y_i \log c_i + (1 - y_i) \log(1 - c_i)] \quad (17)$$

where $y_i = 1$ if and only if query i is matched to a ground truth point, otherwise $y_i = 0$.

Combining all the above terms, the joint training objective function for the CrowdVision module is defined as:

$$\mathcal{L}_{\text{CrowdVision}} = \lambda_1 \mathcal{L}_{\text{count}} + \lambda_2 \mathcal{L}_{\text{loc}} + \lambda_3 \mathcal{L}_{\text{conf}} \quad (18)$$

where λ_1 , λ_2 , and λ_3 are weight coefficients for balancing each loss term. Through this multi-task joint optimization strategy, the CrowdVision module can achieve precise crowd localization while ensuring counting accuracy, providing high-quality foundational data for the subsequent intelligent analysis module.

LLM-Advisor module

The LLM-Advisor module employs large language models as the core analysis engine, integrating crowd counting data with multi-source heterogeneous data in a structured manner, and generating management reports containing problem diagnosis and configuration recommendations through specially designed prompt engineering techniques. As illustrated in Fig. 3, the module consists of four key components: multi-source data integration, prompt engineering framework, LLM analysis engine, and RAG-based knowledge base enhancement.

Multi-source data integration

To enable large language models to perform effective analysis and decision-making, the system needs to integrate multi-source heterogeneous data and construct structured facility profiles. Specifically, input data includes the following four dimensions: (1) **Crowd time-series data**: Real-time headcount statistics and historical usage records from the CrowdVision module, forming time-series features after anomaly detection and smoothing processing; (2) **Geospatial information**: Facility geographic location, POI density of surrounding

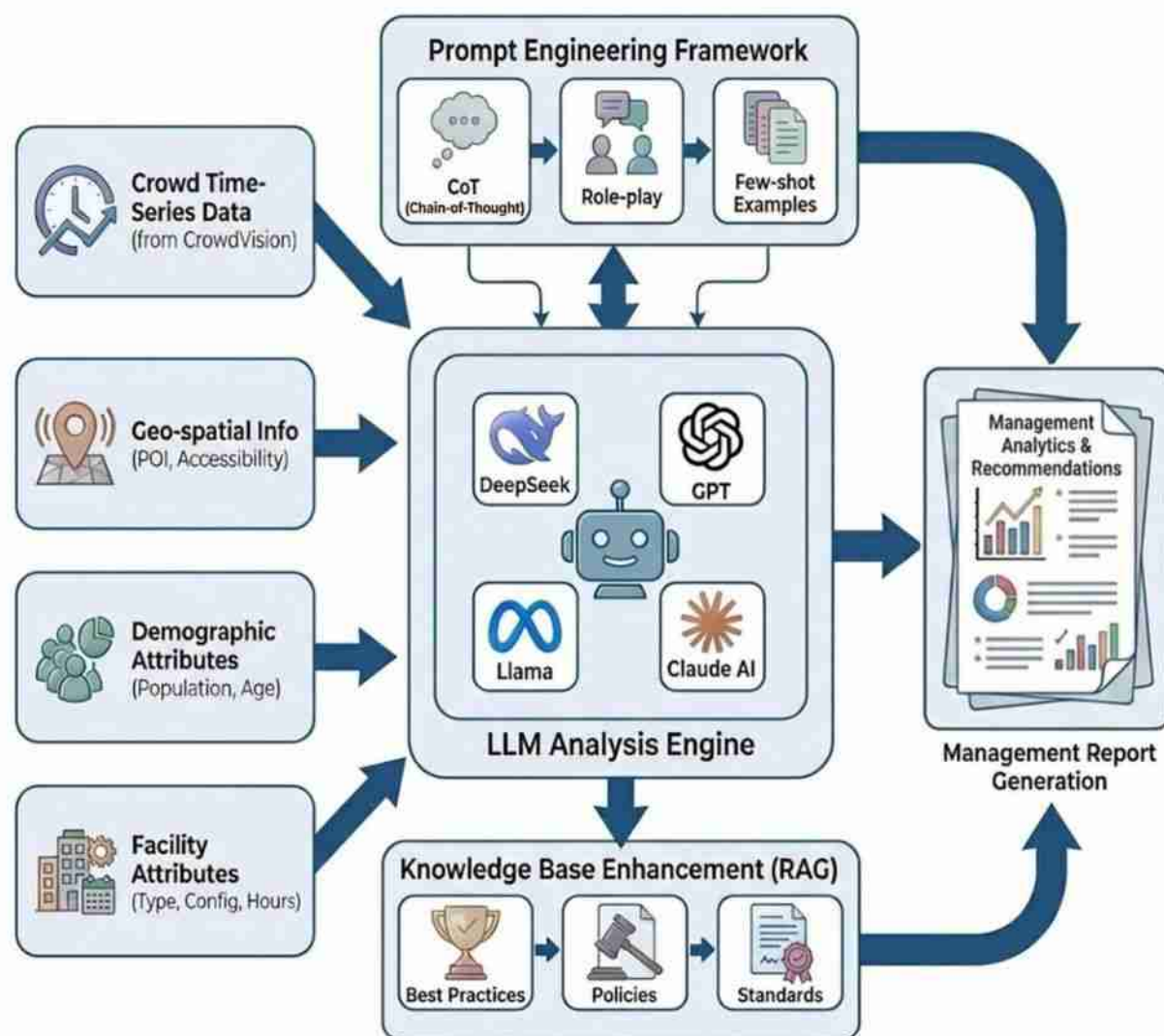


Fig. 3. Architecture of the LLM-Advisor module, showing the integration of four data sources (crowd time-series, geo-spatial information, demographic attributes, and facility attributes) with the prompt engineering framework (CoT, role-play, few-shot examples), multiple LLM backends (DeepSeek, GPT, Llama, Claude), and RAG-based knowledge base enhancement for generating comprehensive management reports.

residential/commercial/transportation facilities, and transportation accessibility indicators; (3) **Demographic characteristics:** Service population size and age structure distribution estimated based on census and mobile signaling data; (4) **Facility attribute information:** Basic attributes such as facility type, site area, equipment configuration, and operating hours. The above multi-dimensional data is standardized and input to the large language model for analysis in the form of structured text.

Prompt engineering design

To guide large language models in generating high-quality analysis reports, we design a multi-stage reasoning prompt framework based on Chain-of-Thought (CoT). This framework integrates role-playing, step-by-step reasoning instructions, structured output specifications, and few-shot examples to guide the model in conducting systematic facility operation analysis. The prompt template is designed as follows:

Prompt Template for Facility Analysis

[Role]

You are a senior public sports facility operations analyst with 15 years of experience in urban planning and facility management. Your expertise includes usage pattern analysis, supply-demand assessment, and operational optimization for public sports venues.

[Task]

Analyze the provided facility data and generate a comprehensive management report. Follow the Chain-of-Thought reasoning process below, completing each step before proceeding to the next.

[Reasoning Steps]

Step 1 - Data Pattern Recognition:

Examine the time-series usage data and identify: (a) daily peak hours and their characteristics, (b) weekly usage patterns (weekday vs. weekend), (c) seasonal trends, (d) any anomalies or outliers. Summarize key statistical features including mean, variance, and periodicity.

Step 2 - Causal Attribution:

Based on patterns identified in Step 1, reason about the underlying causes. Consider factors such as: demographic composition of nearby residents, facility accessibility, competing facilities, weather sensitivity, and local events. Explain why each pattern exists.

Step 3 - Supply-Demand Gap Analysis:

Compare facility capacity against actual demand. Calculate utilization rates for different time slots. Identify temporal and spatial mismatches between supply and demand. Assess whether latent demand exists (e.g., potential users deterred by overcrowding).

Step 4 - Strategy Formulation:

Propose actionable recommendations addressing issues identified in Steps 2-3. For each recommendation: (a) describe the specific action, (b) explain the expected impact, (c) assess implementation feasibility, (d) estimate resource requirements. Prioritize recommendations by cost-effectiveness.

Step 5 - Validation via Counterfactual Reasoning:

For each proposed strategy, conduct counterfactual analysis: "If this strategy had been implemented 6 months ago, what would the expected outcome be?" Identify potential risks and mitigation measures.

[Output Format]

Structure your response as follows:

```
{
  "pattern_analysis": {
    "temporal_patterns": "...",
    "key_statistics": "...",
    "anomalies_detected": "..."
  },
  "causal_factors": [{"factor": "explanation", ...}],
  "supply_demand_assessment": {
    "utilization_rate": "...",
    "gap_severity": "low/medium/high",
    "bottleneck_periods": "..."
  },
  "recommendations": [
    {
      "action": "...",
      "expected_impact": "...",
      "feasibility": "...",
      "priority": "high/medium/low",
    }
  ]
}
```

```

      "confidence_score": 0.0-1.0
    }
  ],
  "risk_analysis": "...
}

```

[Example]

Input: Basketball court in residential area, avg. 45 users/day, peak at 18:00-20:00 (85 users), weekend utilization 95%, weekday afternoon 20%, capacity 50 persons, surrounding population 12,000 (35% youth).

Output:

```

{
  "pattern_analysis": {
    "temporal_patterns": "Strong evening peak (18:00-20:00)
      on weekdays indicating post-work usage; weekend shows
      sustained high usage throughout daytime",
    "key_statistics": "Mean daily users: 45, Peak: 85,
      Weekend/weekday ratio: 2.1",
    "anomalies_detected": "Weekday afternoon significantly
      underutilized despite high youth population"
  },
  "causal_factors": [
    "Evening peak: working adults exercise after work hours",
    "Weekend high usage: families and youth have leisure time",
    "Afternoon gap: youth in school, facility lacks programs
      targeting seniors"
  ],
  "supply_demand_assessment": {
    "utilization_rate": "Weekend peak exceeds capacity by 90%,
      causing overcrowding",
    "gap_severity": "high",
    "bottleneck_periods": "Saturday-Sunday 09:00-18:00,
      Weekday 18:00-20:00"
  },
  "recommendations": [
    {
      "action": "Implement time-slot reservation system to
        distribute weekend demand",
      "expected_impact": "Reduce peak overcrowding by 30%,
        improve user satisfaction",
      "feasibility": "High - low cost, requires only
        software deployment",
      "priority": "high",
      "confidence_score": 0.85
    },
    {
      "action": "Launch senior fitness programs during
        weekday afternoons",
      "expected_impact": "Increase afternoon utilization
        from 20% to 50%",
      "feasibility": "Medium - requires instructor recruitment
        and program design",
      "priority": "medium",
      "confidence_score": 0.72
    }
  ],
  "risk_analysis": "Reservation system may deter casual users;
    mitigation: reserve 20% capacity for walk-ins"
}

```

Now analyze the following facility data:

[INPUT_DATA]

This prompt design uses role-playing to establish a professional analytical tone, guides the model through layer-by-layer reasoning via a five-step chain of thought, and provides complete few-shot examples to help the model understand expected output format and analysis depth. Self-consistency verification further improves output reliability through multiple sampling and consensus.

Knowledge base enhancement

To improve the professionalism and accuracy of analysis, we construct a sports facility management knowledge base as an external knowledge source for Retrieval-Augmented Generation (RAG). The knowledge base contains three types of content: (1) **Best practice case library**: Collects excellent facility management cases from domestic

and international sources, recording problem contexts, measures taken, and implementation effects; (2) **Policy and regulation library**: Integrates policy documents, management specifications, and applicable conditions related to sports facilities; (3) **Technical standards library**: Compiles construction standards, operational specifications, and evaluation indicators for various types of facilities. When generating analysis reports, the system retrieves relevant knowledge entries based on current query semantics and injects them as context into the prompt, making the large language model's output more professional and actionable.

System deployment

Considering the reality that sports facilities are widely distributed with varying network conditions, this system adopts an edge-cloud collaborative deployment architecture. On the edge side, each facility is configured with an embedded computing node responsible for video capture and real-time crowd counting inference, uploading counting results rather than raw video to the cloud, which both reduces bandwidth requirements and ensures privacy security. The CrowdVision model, after quantization and operator optimization, is deployed on edge devices with single-frame inference latency controlled within 50ms. On the cloud side, the LLM-Advisor module integrates data reported from each facility for analysis and decision-making, generates management reports, and distributes them to management terminals. The system uses message queues to implement asynchronous communication between edge nodes and the cloud, ensuring no data loss during network fluctuations. Additionally, the system follows the data minimization principle, with image data processed only on the edge side and only anonymized statistical data stored in the cloud, complying with relevant data protection regulations.

Experiments

Datasets

Crowd counting evaluation datasets

We evaluate the CrowdVision module on two public benchmark datasets: **ShanghaiTech Part B**³⁰ (716 images, averaging 123 persons per image) and **JHU-CROWD++**³¹. The former's medium-to-low density scenarios align well with real sports facility usage; the latter covers diverse scenes including sports venues and parks. We supplement JHU-CROWD++ with sports venue images to construct the **JHU-Sports dataset** (856 images). Evaluation metrics include MAE, RMSE, and Counting Accuracy (CA).

SmartSport-facility dataset

To evaluate the overall efficacy of the SmartSport framework, we construct the **SmartSport-Facility dataset**. For visual data, we select 15 public sports facilities in a city, collecting 52,400 images from March to August 2024, with 6,200 images independently annotated by 3 annotators (Fleiss' Kappa=0.85). We also collect multi-source auxiliary data including geographic location, POI distribution, and meteorological information. Additionally, 12 domain experts conduct blind evaluations of 256 LLM-generated management reports.

Implementation details

CrowdVision module

Network configuration: The backbone network adopts ResNet-18 pretrained on ImageNet, with dilation rates of dilated convolutions set to {2, 4, 8}. The hidden state dimension of the Visual State Space Module (VSSM) is set to 128, and the time step parameter Δ_A is initialized to 0.1. The feature pyramid uniformly outputs 128 channels. The number of point queries N is set to 500, the number of cross-attention layers K is set to 6, with each layer containing 8 attention heads. The confidence threshold τ_{conf} is determined through grid search on the validation set, with a final value of 0.5.

Training strategy: The model is trained using the AdamW optimizer with an initial learning rate of 1×10^{-4} and a weight decay coefficient of 1×10^{-4} . A cosine annealing learning rate schedule is adopted over 300 training epochs. The loss function weights are set as $\lambda_1 = 1.0$ (counting loss), $\lambda_2 = 5.0$ (localization loss), and $\lambda_3 = 2.0$ (confidence loss), with matching cost coefficients $\lambda_{loc} = 2.0$ and $\lambda_{conf} = 1.0$. Data augmentation includes random horizontal flipping, random cropping (crop ratio 0.5–1.0), color jittering (brightness, contrast, saturation variation range ± 0.2), and random grayscale conversion (probability 0.1). The training batch size is 16.

LLM-Advisor module

Model selection: We evaluated five large language models as the analysis engine: GPT-4o, DeepSeek-R1, Qwen-Plus, LLaMA-4, and Claude Opus 4.1. The inference temperature is uniformly set to 0.3 to balance output diversity and stability, with a maximum output token count of 4,096.

Prompt configuration: We adopt the chain-of-thought prompt template described in the previous section, retrieving the top-5 relevant cases from the knowledge base as context for each analysis. To improve output consistency, we employ a self-consistency strategy with 5 sampling iterations per query, determining the final recommendation through majority voting.

Knowledge base construction: The knowledge base is built on FAISS vector indexing, with text embeddings generated using the text-embedding-3-large model (dimension 3,072). The knowledge base contains: 156 best practice cases, 89 policy and regulation documents, and 42 technical standard files, which are chunked into 2,847 knowledge fragments. Retrieval employs cosine similarity matching with a similarity threshold of 0.75.

System integration: Edge nodes communicate with the cloud through Apache Kafka message queues, with message serialization using Protocol Buffers. The cloud service is built on the FastAPI framework, supporting asynchronous request processing. Data storage utilizes InfluxDB (for storing crowd statistics) as a time-series database and PostgreSQL (for storing facility attributes and analysis reports). The system is deployed on Alibaba Cloud, with the LLM-Advisor module configured on a compute instance with 8 CPU cores and 32GB memory.

Method	Year	MAE↓	RMSE↓	CA(%)↑
DM-Count ³²	2020	7.4	11.8	88.2
P2PNet ³³	2021	6.3	9.9	90.1
MAN ³⁴	2022	6.1	9.5	90.5
CLTR ³⁵	2022	6.5	10.6	89.6
STEERER ³⁶	2023	5.8	8.5	91.2
APGCC ³⁷	2024	5.5	8.2	91.8
PET ³⁸	2024	5.6	8.4	91.5
CrowdVision (ours)	—	5.2	7.8	92.4

Table 1. Performance comparison on ShanghaiTech Part B dataset.

Method	Year	MAE↓	RMSE↓	CA(%)↑
DM-Count ³²	2020	12.6	19.3	85.1
P2PNet ³³	2021	11.2	17.8	87.2
MAN ³⁴	2022	10.8	16.9	88.0
CLTR ³⁵	2022	11.5	18.2	86.8
STEERER ³⁶	2023	10.3	15.7	89.1
APGCC ³⁷	2024	9.8	15.1	90.2
PET ³⁸	2024	10.1	15.4	89.6
CrowdVision (ours)	—	9.1	14.2	91.5

Table 2. Performance comparison on JHU-Sports dataset.

Method	MAE↓	RMSE↓	CA(%)↑
DM-Count ³²	1.12	1.85	85.4
P2PNet ³³	0.95	1.62	87.6
MAN ³⁴	0.88	1.51	88.5
STEERER ³⁶	0.76	1.28	90.2
APGCC ³⁷	0.68	1.15	91.2
CrowdVision (ours)	0.58	0.96	93.8

Table 3. Performance comparison on SmartSport-Facility dataset.

Main experimental results and analysis

Crowd counting performance evaluation

We evaluate the performance of the CrowdVision module on the ShanghaiTech Part B and JHU-Sports datasets, comparing against representative methods from 2020–2024.

As shown in Tables 1 and 2, CrowdVision achieves the best performance on both datasets. On ShanghaiTech Part B, CrowdVision achieves an MAE of 5.2 and a CA of 92.4%; on JHU-Sports, it achieves an MAE of 9.1 and a CA of 91.5%. Compared to the second-best method APGCC, the MAE is reduced by 5.5% and 7.1% on the two datasets, respectively. This advantage is primarily attributed to: (1) the Visual State Space Module enabling efficient global context modeling through its selective mechanism; (2) the cross-directional scanning strategy aggregating spatial information from four directions; and (3) the point query mechanism decomposing the counting task into independent point predictions, avoiding the Gaussian kernel overlap problem in density map regression.

Furthermore, we validate the model’s practical application effectiveness on the self-constructed SmartSport-Facility dataset, with results shown in Table 3.

This dataset features scenarios closer to real sports facility applications with moderate crowd density (averaging 7.7 persons per image). CrowdVision achieves a CA of 93.8%, consistent with the results reported in the abstract, demonstrating that the model possesses good generalization capability in practical sports facility scenarios.

Computational efficiency analysis

To evaluate the practical deployment feasibility, we compare the computational efficiency of CrowdVision with representative methods. All experiments are conducted on a single NVIDIA RTX 3090 GPU with input resolution of 512 × 512. Table 4 presents the comparison results.

Method	Backbone	Params (M)	FLOPs (G)	Inference (ms)	FPS
P2PNet	VGG16-bn	21.8	94.5	35.8	27.9
MAN	VGG-19	30.9	105.8	42.5	23.5
STEERER	HRNet-48	65.2	156.4	58.6	17.1
APGCC	VGG-16	25.4	98.2	37.6	26.6
CrowdVision	ResNet-18	18.6	25.8	17.8	56.2

Table 4. Computational efficiency comparison.

Model	Info. Comp.	Anal. Logic	Rec. Feas.	Prof. Stand.	Average
LLaMA-4	3.58	3.45	3.32	3.53	3.47
Qwen-Plus	3.85	3.78	3.65	3.80	3.77
DeepSeek-R1	4.08	4.15	3.92	4.05	4.05
GPT-4o	4.22	4.18	4.05	4.11	4.14
Claude Opus 4.1	4.28	4.25	4.12	4.15	4.20

Table 5. Human evaluation results of different large language models (full score: 5).

As shown in Table 4, CrowdVision achieves the best efficiency across all metrics. The model has the smallest parameter count (18.6M) due to the lightweight ResNet-18 backbone, and the lowest computational cost (25.8 GFLOPs) attributed to both the efficient backbone architecture and the linear complexity of the Visual State Space Module. The inference speed of 56.2 FPS on RTX 3090 significantly outperforms VGG-based methods (23–28 FPS) and HRNet-based STEERER (17.1 FPS). After INT8 quantization and TensorRT optimization, the model achieves an inference latency of 42.5ms (23.5 FPS) on edge devices (NVIDIA Jetson AGX Orin), well within the 50ms real-time requirement for practical sports facility monitoring.

LLM-Advisor module evaluation

Table 5 presents the human evaluation results of different large language models on the management report generation task. Twelve domain experts conducted blind evaluations of 256 reports on a 5-point scale across four dimensions.

As shown in Table 5, Claude Opus 4.1 achieves the highest scores across all evaluation dimensions, with an average score of 4.20. GPT-4o ranks second with 4.14 points, performing close to Claude Opus 4.1 in information completeness and analytical logic. DeepSeek-R1, as a domestic model, demonstrates outstanding performance, particularly achieving 4.15 in the analytical logic dimension, exhibiting strong reasoning capabilities. Among open-source models, Qwen-Plus outperforms LLaMA-4 overall but still shows a notable gap compared to closed-source models.

Further analysis of scores across dimensions reveals that “Recommendation Feasibility” is the lowest-scoring dimension for all models, reflecting that large language models still have room for improvement in generating specific actionable recommendations. This may be related to the models’ lack of specialized knowledge in the sports facility management domain. After introducing the RAG knowledge base, the average improvement across all models in the “Professional Standardization” dimension is 0.35 points, validating the effectiveness of domain knowledge injection in enhancing output professionalism. Considering both model performance and inference cost, this framework adopts Claude Opus 4.1 as the default analysis engine.

To ensure evaluation quality, we implemented a two-stage quality control mechanism for expert ratings. In the first stage, statistical anomaly detection identified ratings deviating more than $2 \times \text{MAD}$ (Median Absolute Deviation) from the median for each report-dimension pair. In the second stage, flagged ratings underwent LLM-assisted semantic review to detect content-rating inconsistencies and systematic biases, followed by manual confirmation by the research team. Approximately 6.5% of ratings were excluded as anomalous, while all 12 experts retained valid ratings in the final analysis.

Based on the filtered data, we assessed inter-expert consistency using established reliability metrics. The overall Krippendorff’s $\alpha = 0.76$ indicates substantial agreement according to standard guidelines. The Intraclass Correlation Coefficient ICC(2,k) values for each dimension were: Information Completeness (0.81), Analytical Logic (0.78), Recommendation Feasibility (0.73), and Professional Standardization (0.79), all indicating good inter-rater reliability. The relatively lower ICC for Recommendation Feasibility reflects the inherently subjective nature of assessing actionability across experts with different management backgrounds.

Ablation studies

To verify the effectiveness of key components in the SmartSport framework, we conduct ablation experiments on both the CrowdVision module and the LLM-Advisor module.

CrowdVision module ablation

Table 6 presents the ablation results for components of the CrowdVision module, conducted on the SmartSport-Facility dataset.

Config.	VSSM	Cross-Dir. Scan	Feat. pyramid	Scale-Adapt.	Point query	MAE↓	RMSE↓	CA↑
Baseline	×	×	×	×	×	1.25	1.82	83.6
V1	×	×	✓	×	×	0.95	1.48	87.5
V2	✓	×	✓	×	×	0.78	1.22	90.2
V3	✓	✓	✓	×	×	0.72	1.15	91.4
V4	✓	✓	✓	✓	×	0.68	1.08	92.1
CrowdVision	✓	✓	✓	✓	✓	0.58	0.96	93.8

Table 6. Ablation results for CrowdVision module.

Config.	RAG KB	CoT Prompt	Multi-Source	Self-Consist.	Role Setting	Info. Comp.	Anal. Logic	Rec. Feas.	Prof. Stand.	Avg.
Baseline	×	×	×	×	×	3.45	3.28	3.15	3.32	3.30
A1	×	×	✓	×	×	3.72	3.55	3.48	3.58	3.58
A2	×	✓	✓	×	×	3.88	3.82	3.65	3.72	3.77
A3	✓	✓	✓	×	×	4.05	4.02	3.88	3.95	3.98
A4	✓	✓	✓	✓	×	4.15	4.12	3.98	4.02	4.07
LLM-Advisor	✓	✓	✓	✓	✓	4.28	4.25	4.12	4.15	4.20

Table 7. Ablation results for LLM-Advisor module (full score: 5).

As shown in Table 6, each component contributes positively to model performance: (1) Adding the feature pyramid to the Baseline (V1) improves CA from 83.6% to 87.5%, indicating that multi-scale feature fusion is crucial for handling crowds at different distances; (2) Introducing the Visual State Space Module VSSM (V2) improves CA by 2.7 percentage points, demonstrating VSSM’s advantage in global context modeling; (3) Adopting the cross-directional scanning strategy (V3) further improves CA to 91.4%, validating the effectiveness of four-directional scanning in capturing spatial dependencies; (4) Adding the scale-adaptive attention mechanism (V4) improves CA by 0.7 percentage points, indicating that dynamic adjustment of level weights helps adapt to different scenarios; (5) The complete CrowdVision model with the point query mechanism achieves a CA of 93.8%, validating the advantage of the point query paradigm in avoiding Gaussian kernel overlap issues.

LLM-Advisor module ablation

Table 7 presents the ablation results for components of the LLM-Advisor module, using Claude Opus 4.1 as the base model.

As shown in Table 7, each component significantly contributes to the performance improvement of the LLM-Advisor module. Multi-source data integration serves as the foundation for enhancing report quality, improving the “Recommendation Feasibility” dimension by 0.33 points, indicating that auxiliary information such as geographic location and population distribution helps the model generate more targeted management recommendations. The introduction of chain-of-thought prompting shows the most significant improvement in “Analytical Logic” (+0.27 points), validating the effectiveness of structured reasoning frameworks in guiding the model to perform deeper analysis. The addition of the RAG knowledge base improves “Professional Standardization” by 0.23 points, demonstrating that the retrieval-augmented mechanism can effectively compensate for the lack of domain-specific knowledge in general-purpose large models. The self-consistency sampling strategy improves output stability and reliability through a multi-sampling voting mechanism. Role setting, as an important component of prompt engineering, further strengthens the model’s professional analysis capabilities, bringing the complete framework’s average score to 4.20.

Hyperparameter sensitivity analysis

To gain deeper insights into how key hyperparameters affect the performance of the CrowdVision module, we conduct sensitivity analysis on three core hyperparameters using the SmartSport-Facility dataset: the number of point queries N , the number of cross-attention layers K , and the confidence threshold τ_{conf} . The experimental results are shown in Fig. 4.

From the figure, we observe the following: (1) For the number of point queries N , performance improves as N increases from 100 to 500, reaching optimal performance at $N = 500$ (MAE=0.58, CA=93.8%). Further increasing N leads to slight performance degradation, likely due to redundant predictions introduced by excessive query points. (2) For cross-attention layers K , performance continuously improves in the range of 2-6, with $K = 6$ achieving optimal results. Further increasing the number of layers yields limited benefits while significantly increasing computational cost. (3) The confidence threshold τ_{conf} has a significant impact on model performance: a threshold that is too low (0.3-0.4) retains too many low-confidence predictions leading to false positives, while a threshold that is too high (0.6-0.8) filters out some correct predictions leading to false negatives. $\tau_{conf} = 0.5$ represents the optimal balance point. Overall, CrowdVision maintains stable high performance across a wide range of hyperparameter settings, demonstrating good robustness to hyperparameter selection.

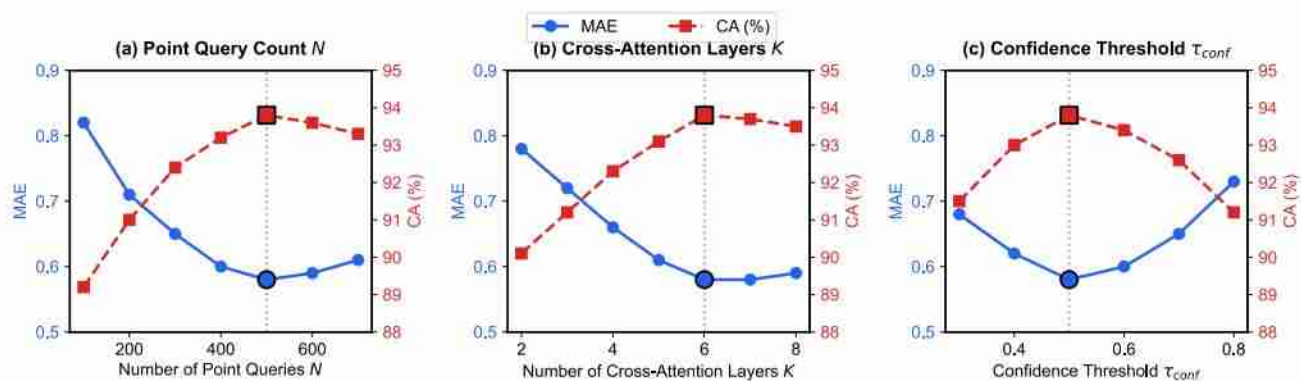


Fig. 4. Hyperparameter sensitivity analysis. (a) Number of point queries N ; (b) Number of cross-attention layers K ; (c) Confidence threshold τ_{conf} . Blue solid lines represent MAE, red dashed lines represent CA, and black circles mark the optimal configuration points.

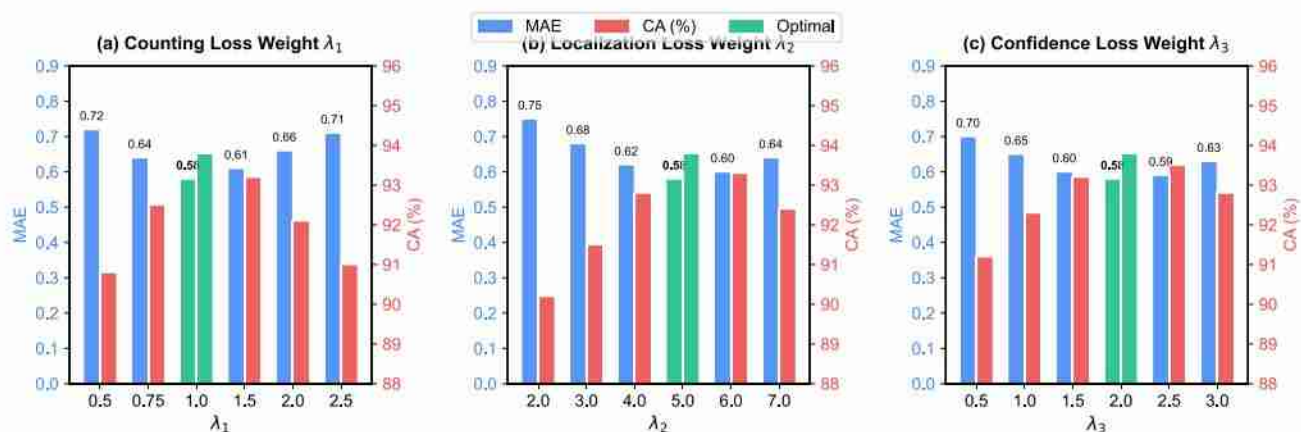


Fig. 5. Loss function weight sensitivity analysis. (a) Counting loss weight λ_1 ; (b) Localization loss weight λ_2 ; (c) Confidence loss weight λ_3 . Blue bars represent MAE, red bars represent CA, and green bars indicate optimal configurations.

Furthermore, we analyze the impact of loss function weights on model performance, including the counting loss weight λ_1 , localization loss weight λ_2 , and confidence loss weight λ_3 . The experimental results are shown in Fig. 5.

From the figure, we observe the following: (1) A counting loss weight λ_1 that is too small leads to insufficient counting constraints, while a value that is too large suppresses the learning of position prediction; $\lambda_1 = 1.0$ represents the optimal balance point. (2) The localization loss weight λ_2 has the most significant impact on model performance, with optimal performance achieved at $\lambda_2 = 5.0$, indicating that precise position prediction is crucial for crowd counting tasks. (3) The confidence loss weight achieves optimal performance at $\lambda_3 = 2.0$; an appropriate confidence constraint helps the model learn to distinguish foreground points from background points. The optimal configuration of the three loss weights is $\lambda_1 = 1.0$, $\lambda_2 = 5.0$, $\lambda_3 = 2.0$, consistent with the settings described in the implementation details.

Visualization results

To intuitively demonstrate the crowd counting and localization capabilities of the CrowdVision module, we present visualization results on representative scenarios from the SmartSport-Facility dataset. Figure 6 shows the detection results across four different scenes, where green dots indicate the predicted person locations.

As shown in Fig. 6, the CrowdVision module demonstrates robust performance across diverse and challenging scenarios: (1) In the urban park scene (a), the model successfully detects individuals distributed across a wide area with varying distances from the camera, demonstrating effective multi-scale perception capability. (2) The street gathering scene (b) presents a dense crowd in a circular formation, where the model accurately localizes individuals even in areas of high density, benefiting from the point query mechanism that avoids Gaussian kernel overlap issues. (3) The nighttime plaza scene (c) showcases the model's robustness to low-light conditions, maintaining accurate detection despite challenging illumination. (4) The outdoor festival scene (d) contains complex backgrounds with colorful tents and structures, yet the model effectively distinguishes persons from



Fig. 6. Visualization of crowd counting results on different scenarios. (a) Urban park with moderate crowd density; (b) Street gathering event with dense crowd; (c) Nighttime plaza scene; (d) Outdoor festival with complex background. Green dots represent detected person locations. The count statistics are displayed in the upper-left corner of each image.

background clutter. These visualization results validate that the proposed CrowdVision module can reliably perform crowd counting and localization in real-world sports facility and public venue scenarios.

Broader application prospects

The proposed LLM-Advisor module exhibits strong conceptual connections with modern context-aware recommendation systems (CARS). Similar to dynamic intention-aware recommendation frameworks³⁹ that model users' preferences based on contextual features and item attributes, our LLM-Advisor incorporates real-time environmental context (crowd density, temporal patterns, weather conditions) and structured facility attributes through the RAG knowledge base to generate situation-appropriate management recommendations. This design pattern aligns with recent advances in hierarchical attention networks for context-aware user preference modeling⁴⁰, group attention mechanisms for attribute-aware collaborative filtering⁴¹, and positional encoding techniques for capturing temporal dependencies in sequential recommendations⁴².

This conceptual alignment suggests several promising directions for extending the SmartSport framework. First, the LLM-Advisor architecture can be generalized to other smart city scenarios requiring context-aware decision support, such as traffic management, public event coordination, and emergency response systems. Second, integrating collaborative filtering mechanisms could enable cross-facility learning, where successful management strategies from similar facilities inform recommendations for new scenarios. Third, the attribute-aware design pattern can be extended to incorporate user-specific preferences, enabling personalized facility recommendations based on individual activity patterns.

Limitations and future work

Limitations The current SmartSport framework adopts a two-stage design: the first stage employs the CrowdVision module for crowd counting, while the second stage uses the LLM-Advisor module for management report generation. Although this decoupled design reduces system complexity, it exhibits notable limitations. On one hand, CrowdVision only transmits numerical counting results to subsequent modules, while the rich semantic information contained in the original images (such as crowd density distribution, activity types, and facility occupancy conditions) is not fully utilized, resulting in information loss. On the other hand, errors between the two stages accumulate progressively, with counting deviations in the first stage directly affecting the accuracy of the second stage analysis. Furthermore, the CrowdVision module experiences decreased counting precision under extreme lighting conditions and severe occlusion scenarios, and only outputs headcount statistics without fine-grained understanding of crowd behavior patterns, limiting the depth of subsequent analysis. Moreover, the current pipeline does not explicitly propagate estimation uncertainty from the CrowdVision module to the LLM-Advisor. Future work will explore uncertainty-aware interfaces, such as providing confidence intervals alongside point estimates, to enable the LLM-Advisor to generate more calibrated recommendations when counting reliability varies.

Future work To address the aforementioned limitations, future work will focus on exploring end-to-end architectures based on multimodal large models. Specifically, vision-language models (such as GPT-4o and Qwen-VL) can be leveraged to directly process facility surveillance images, simultaneously accomplishing scene

understanding, crowd analysis, and report generation within a single model, achieving end-to-end mapping from raw visual input to management recommendations. This architecture can preserve complete semantic information from images, avoid the information loss and error accumulation inherent in two-stage designs, while simplifying system deployment and maintenance workflows. Additionally, future research will explore video understanding models to capture temporal dynamic features, as well as multi-agent collaboration mechanisms to achieve regional-level facility resource coordination and allocation.

Conclusion

This study addresses the challenges of low utilization monitoring efficiency and insufficient decision-making intelligence in public sports facility management by proposing SmartSport, an intelligent management framework integrating computer vision and large language models. The framework comprises two core modules: the CrowdVision module employs a lightweight network based on visual state space models to achieve automated monitoring of facility usage, utilizing cross-directional scanning strategies and point query mechanisms to achieve precise crowd counting and localization while maintaining computational efficiency; the LLM-Advisor module employs large language models as the analysis engine, integrating crowd time-series data with multi-source geographic information, and generating professional management reports through chain-of-thought prompting and RAG knowledge base enhancement. Experimental results demonstrate that the CrowdVision module achieves a counting accuracy of 93.8% on the self-constructed SmartSport-Facility dataset, with performance superior to existing methods on public benchmark datasets; management recommendations generated by the LLM-Advisor module received a rating of 4.2 out of 5 points through blind evaluation by 12 domain experts.

Data availability

No datasets were generated or analysed during the current study.

Code availability

The source code for the SmartSport framework, including the CrowdVision crowd counting module and the LLM-Advisor prompt templates, is publicly available on Zenodo at <https://doi.org/10.5281/zenodo.18721457>. The repository contains model training and inference scripts, data preprocessing pipelines, and configuration files necessary to reproduce the main results reported in this paper.

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Author contributions

Pingping Li conceived the study, developed the methodology, performed the investigation, curated the data, and wrote the original draft. Churuo Xu supervised the project, managed resources, secured funding, and revised the manuscript. All authors have read and approved the final manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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